

DAY: _____

DATE: _____

Linear Simple $\hat{Y}$ regression model:

$$Y = \alpha + \beta X + E$$

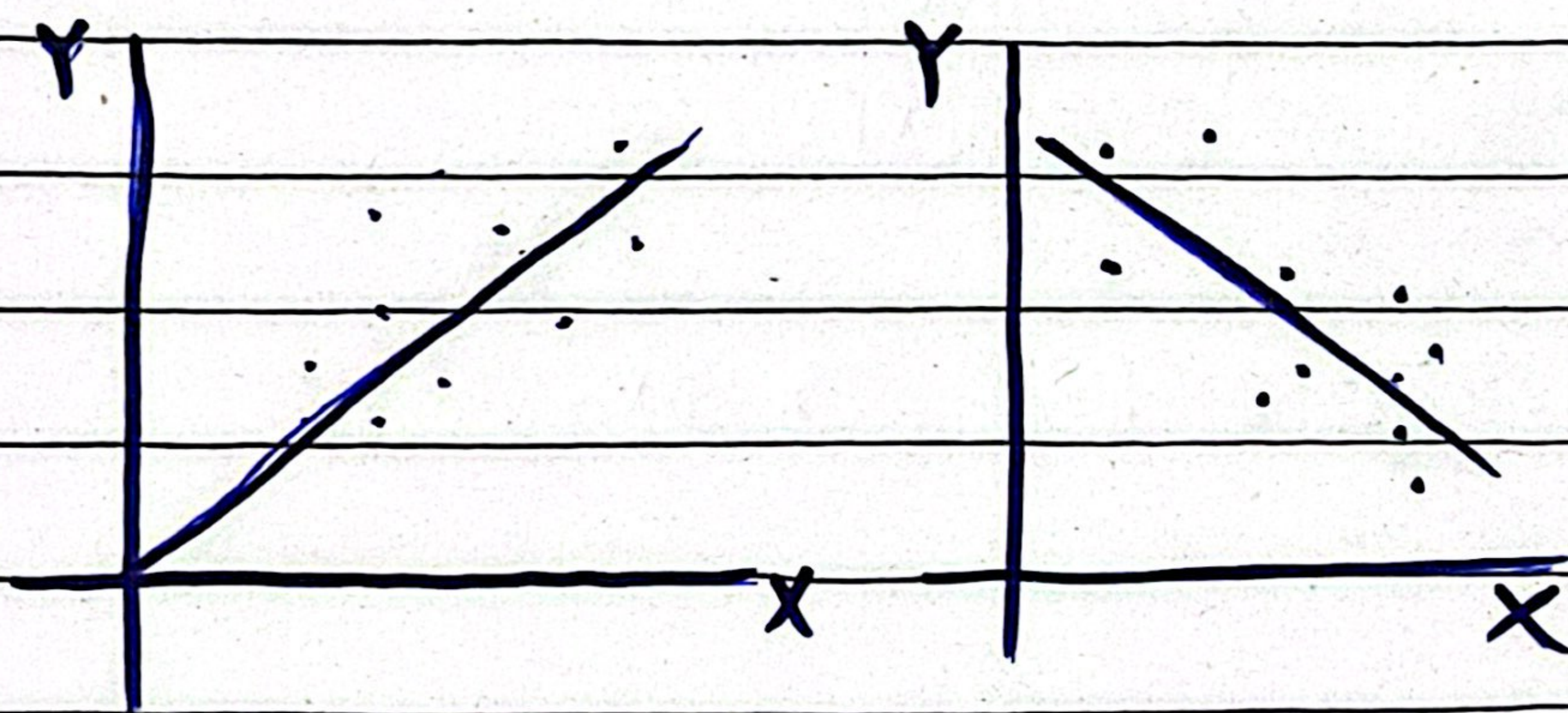
Here,

Y = Dependent Variable. (Explained Variable)

X = Independent Variable. (Explanatory Variable)

α and β are the parameters of the model

E = Error term.



Here α is called Y-intercept

β is called the slope of the line. or regression coefficient.

Ratio of ΔX and ΔY is called slope.

Estimated simple regression line is $\hat{Y} = a + bX$.

Minimize the effect of error.

DAY: _____

DATE: _____

Here;

$$b = \frac{\sum (x - \bar{x})(y - \bar{y})}{\sum (x - \bar{x})^2} = \frac{n \sum x^2 - \sum x \cdot \sum y}{n[\sum x^2 - (\sum x)^2]}$$

① Co-variation is there in the numerator.

② Co-variation divided by the variance of the X.

$$= \frac{\sum x^2 - n \cdot \bar{x} \cdot \bar{y}}{\sum x^2 - n(\bar{x})^2}$$

$$a = \frac{\sum y + b \sum x}{n} = \bar{y} - b\bar{x}$$